

DESKTOP PROCESSORS

We await the the promising successor next year

There's a sense of déjà vu right now, the events that have led up to now remind us of AMD's Bulldozer architecture which was pretty much the talk of the town. Bulldozer didn't deliver but AMD's Zen Architecture seems to be showing promise. AMD's Zen will be coming to the consumer segment as Summit Ridge and for the enterprise segment as Naples. The former has already been demonstrated in August this year when AMD officials pitted an 8-core Zen CPU against Intel's Core i7 6900K which is also an 8-core CPU. Both of which were clocked at 3.0 GHz and tested with a Blender render. The AMD CPU did perform just as well as the Intel CPU and if the final batch scales well with clock speed then we're going to see a 'cores race' in the CPU segment. AMD has even revealed that Zen makes a tremendous leap in IPC performance over the previous architecture. How much of an increment? Over 40 per cent apparently. We still need to know the rest of the parameters around which these numbers were obtained to get a better perspective.

Intel on the other side is ramping up to meet this threat head-on. Around CES we should see Intel releasing their Kaby Lake desktop lineup which will reportedly feature higher clock speed, an improved graphics core and support Intel Optane technology. Optane is probably going to be ridiculously expensive with an equally high performance to match.

ZERO 1 WINNER

Intel® Core™ i7-6950X Extreme Edition

Unsurprisingly, the Core i7-6950X Extreme Edition takes the Zero1 award continuing a trend that's been going on for quite a few years purely because of the lack of competition. While AMD has been hinting at equalising Broadwell-E's performance with Zen, it does stand to reason that by the time Summit Ridge does release, Intel might have something up its sleeve to combat AMD's offering. The Core i7-6950X offers 10 cores / 20 threads which is a lot more than what most mainstream users can make do with. Moreover, the 6950X is way more expensive than previous enthusiasts flagships at \$1723 or ₹1,30,000 which makes owning a 2011-v3 rig a lot more expensive. However, this is the end of the road for the X99 platform as X299 is slated



Intel® Core™
i7-6950X Extreme
Edition

Price: ₹1,30,000



Intel® Core™
i7-6900K
Price: ₹81,500



Intel® Core™
i7-6850K
Price: ₹48,250



Intel® Core™
i5-6500
Price: ₹14,850

to take its place along with Skylake-X around August-November 2017 when the X99 platform completes three years.

RUNNER-UPS

Intel® Core™ i7-6900K and Intel® Core™ i7-6850K

With more SKU's in the enthusiast segment it's quite obvious that the runner-ups are also LGA 2011-v3 CPUs. And as is the case with SKUs in the high-end segment the price difference between the 6900K and the 6950X amount to approximately ₹50,000 but the performance difference doesn't scale accordingly. The same can be said for the performance difference between the 6900K and the 6850K. And unlike last year's top three SKUs, where there were differences in the number of PCI Express lanes available per SKU, we have the same configuration across all three of this year's winners. So in a way, the complexity of choosing between the three SKUs has considerably reduced as it becomes a simple question of getting more IPC.

BEST BUY

Intel® Core™ i5-6500

It's a tough choice between the Intel 6402P and the 6500 in the budget segment. Both have four cores and four threads but the significance of the 'P' suffix hasn't been properly explained on Intel's web site. Historically, the 'P' was used to identify a disabled IGP but the 6402P does have an Intel HD Graphics 510 while the 6400 without the suffix has an Intel HD Graphics 530. While the 6402P may be better clocked, the very fact that it has comparatively poorer IGP breaks it for us. Which is why we'd prefer the i5 6500. **d**